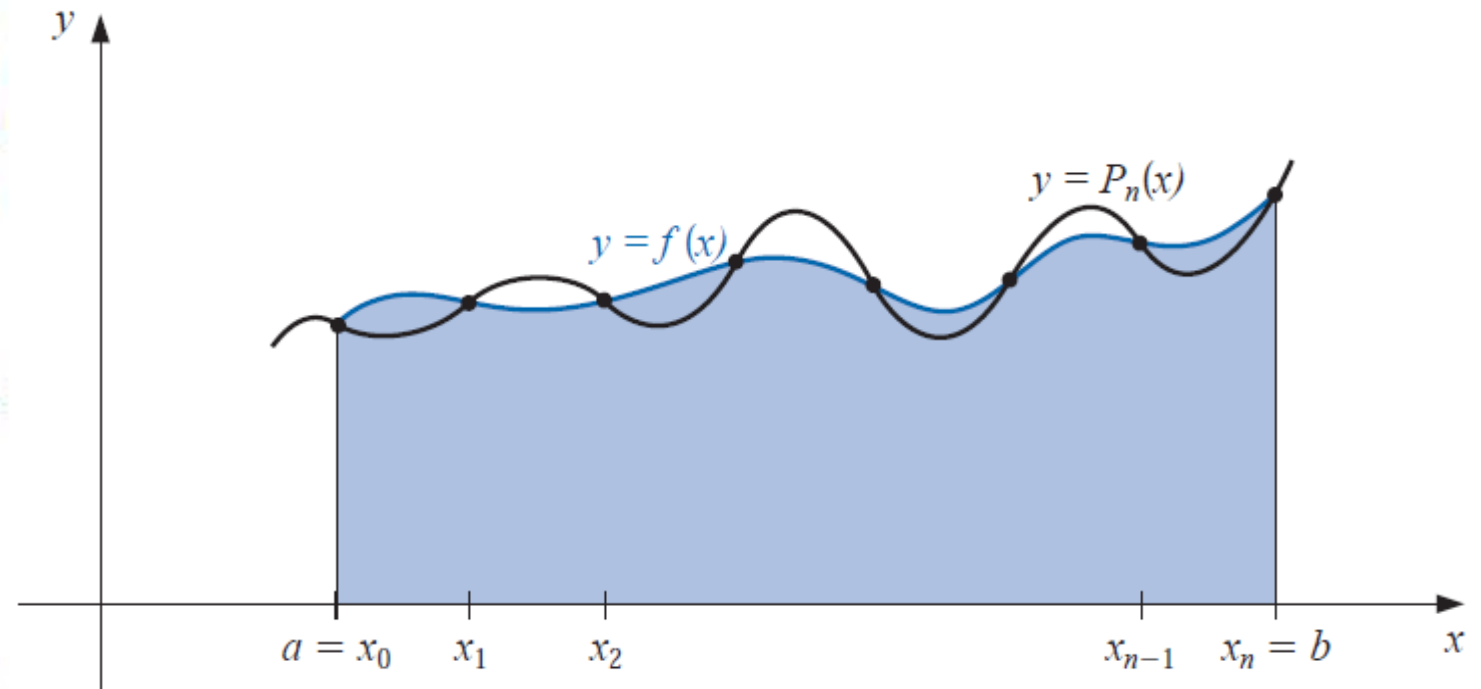


The open and closed terminology for methods implies that the open methods use as nodes only points in the open interval, (a, b) to approximate $\int_a^b f(x) dx$. The closed methods include the points a and b of the closed interval $[a, b]$ as nodes.



Closed Newton-Cotes Formulas

Kapali Newton (ode)

The $(n+1)$ -point closed Newton-Cotes formula uses nodes $x_i = x_0 + ih$, for $i = 0, 1, \dots, n$, where $x_0 = a$, $x_n = b$ and $h = (b - a)/n$. (See Figure 4.5.) It is called closed because the endpoints of the closed interval $[a, b]$ are included as nodes.

uc noktalar dahil!

The formula assumes the form

$$\int_a^b f(x) dx \approx \sum_{i=0}^n a_i f(x_i),$$

where

$$a_i = \int_{x_0}^{x_n} L_i(x) dx = \int_{x_0}^{x_n} \prod_{\substack{j=0 \\ j \neq i}}^n \frac{(x - x_j)}{(x_i - x_j)} dx.$$

$$I = \int_a^b P_n(x) dx = \sum_{i=0}^n \left[f(x_i) \int_a^b \ell_i(x) dx \right] = \sum_{i=0}^n A_i f(x_i) \quad (6.2a)$$

where

$$A_i = \int_a^b \ell_i(x) dx, \quad i = 0, 1, \dots, n \quad (6.2b)$$

Equations (6.2) are the *Newton-Cotes formulas*. Classical examples of these formulas are the *trapezoidal rule* ($n = 1$), *Simpson's rule* ($n = 2$), and *3/8 Simpson's rule* ($n = 3$).

Some of the common **closed Newton-Cotes formulas** with their error terms are listed. Note that in each case the unknown value ξ lies in (a, b) .

$n = 1$: Trapezoidal rule (Yamuk kuralı)

$$\int_{x_0}^{x_1} f(x) dx = \frac{h}{2}[f(x_0) + f(x_1)] - \frac{h^3}{12}f''(\xi), \quad \text{where } x_0 < \xi < x_1. \quad (4.25)$$

$n = 2$: Simpson's rule (Simpson kuralı)

$$\int_{x_0}^{x_2} f(x) dx = \frac{h}{3}[f(x_0) + 4f(x_1) + f(x_2)] - \frac{h^5}{90}f^{(4)}(\xi), \quad \text{where } x_0 < \xi < x_2. \quad (4.26)$$

$n = 3$: Simpson's Three-Eighths rule (3/8 Simpson kuralı)

$$\int_{x_0}^{x_3} f(x) dx = \frac{3h}{8}[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)] - \frac{3h^5}{80}f^{(4)}(\xi), \quad (4.27)$$

where $x_0 < \xi < x_3$.

Open Newton-Cotes Formulas

(Açık Newton Cotes)

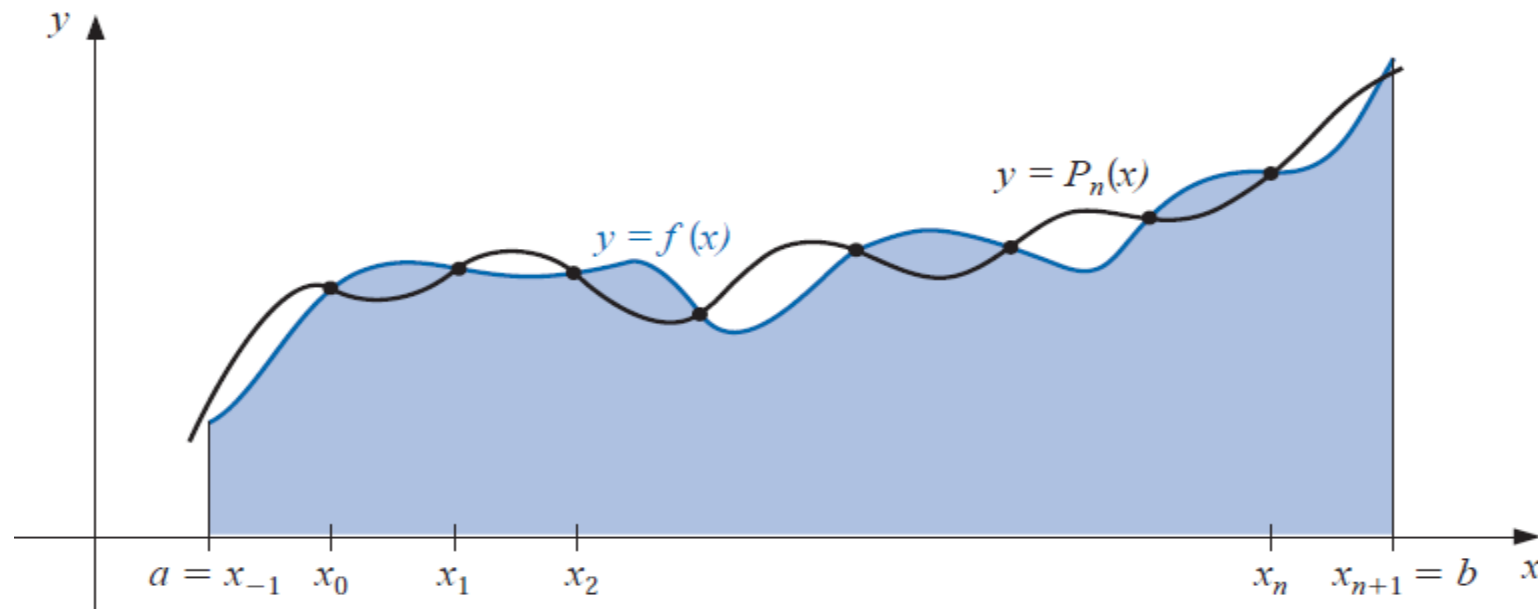
The *open Newton-Cotes formulas* do not include the endpoints of $[a, b]$ as nodes. They use the nodes $x_i = x_0 + ih$, for each $i = 0, 1, \dots, n$, where $h = (b - a)/(n + 2)$ and $x_0 = a + h$. This implies that $x_n = b - h$, so we label the endpoints by setting $x_{-1} = a$ and $x_{n+1} = b$, as shown in Figure 4.6 on page 200. Open formulas contain all the nodes used for the approximation within the open interval (a, b) . The formulas become

$$\int_a^b f(x) dx = \int_{x_{-1}}^{x_{n+1}} f(x) dx \approx \sum_{i=0}^n a_i f(x_i),$$

where

$$a_i = \int_a^b L_i(x) dx.$$

↓
uç noktalar
dahil değil!



Some of the common **open Newton-Cotes** formulas with their error terms are as follows:

$n = 0$: Midpoint rule *(Orta Nokta)*

$$\int_{x_{-1}}^{x_1} f(x) dx = 2h f(x_0) + \frac{h^3}{3} f''(\xi), \quad \text{where } x_{-1} < \xi < x_1. \quad (4.29)$$

$n = 1$:

$$\int_{x_{-1}}^{x_2} f(x) dx = \frac{3h}{2} [f(x_0) + f(x_1)] + \frac{3h^3}{4} f''(\xi), \quad \text{where } x_{-1} < \xi < x_2. \quad (4.30)$$

$n = 2$:

$$\int_{x_{-1}}^{x_3} f(x) dx = \frac{4h}{3} [2f(x_0) - f(x_1) + 2f(x_2)] + \frac{14h^5}{45} f^{(4)}(\xi), \quad (4.31)$$

where $x_{-1} < \xi < x_3$.

Example 2 Compare the results of the closed and open Newton-Cotes formulas listed as (4.25)–(4.28) and (4.29)–(4.32) when approximating

$$\int_0^{\pi/4} \sin x \, dx = 1 - \sqrt{2}/2 \approx 0.29289322.$$

Solution For the closed formulas we have

$$n = 1 : \quad \frac{(\pi/4)}{2} \left[\sin 0 + \sin \frac{\pi}{4} \right] \approx 0.27768018 \qquad \int_{x_0}^{x_1} f(x) \, dx = \frac{h}{2} [f(x_0) + f(x_1)]$$

$$n = 2 : \quad \frac{(\pi/8)}{3} \left[\sin 0 + 4 \sin \frac{\pi}{8} + \sin \frac{\pi}{4} \right] \approx 0.29293264 \qquad \int_{x_0}^{x_2} f(x) \, dx = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)]$$

$$n = 3 : \quad \frac{3(\pi/12)}{8} \left[\sin 0 + 3 \sin \frac{\pi}{12} + 3 \sin \frac{\pi}{6} + \sin \frac{\pi}{4} \right] \approx 0.29291070$$

$$\int_{x_0}^{x_3} f(x) \, dx = \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)]$$

and for the open formulas we have

$$n = 0 : \quad 2(\pi/8) \left[\sin \frac{\pi}{8} \right] \approx 0.30055887$$

$$n = 1 : \quad \frac{3(\pi/12)}{2} \left[\sin \frac{\pi}{12} + \sin \frac{\pi}{6} \right] \approx 0.29798754$$

$$n = 2 : \quad \frac{4(\pi/16)}{3} \left[2 \sin \frac{\pi}{16} - \sin \frac{\pi}{8} + 2 \sin \frac{3\pi}{16} \right] \approx 0.29285866$$

$n = 0$: Midpoint rule

$$\int_{x_{-1}}^{x_1} f(x) \, dx = 2h f(x_0) + \frac{h^3}{3} f''(\xi),$$

$n = 1$:

$$\int_{x_{-1}}^{x_2} f(x) \, dx = \frac{3h}{2} [f(x_0) + f(x_1)] .$$

$n = 2$:

$$\int_{x_{-1}}^{x_3} f(x) \, dx = \frac{4h}{3} [2f(x_0) - f(x_1) + 2f(x_2)]$$

n	0	1	2	3	4
Closed formulas		0.27768018	0.29293264	0.29291070	0.29289318
Error		0.01521303	0.00003942	0.00001748	0.00000004
Open formulas	0.30055887	0.29798754	0.29285866	0.29286923	
Error	0.00766565	0.00509432	0.00003456	0.00002399	